



CRMs DataSpace

RFCS PROJECT

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Materials for the Green Deal: Recovery
Potential from Closed Extractive Coal Waste
Facilities

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Deliverable 4.1: Governance framework



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Executive Summary

This deliverable presents the governance framework for the Common European Data Space on Critical Raw Materials, developed under WP4 of the CRMsDS project. Its main objective is to define the organisational, legal and operational foundations required to establish and operate a Minimum Viable Data Space (MVDS) aligned with European Data Space principles, while respecting data sovereignty, interoperability and trust.

This framework provides a structured set of rules, roles and processes that enable independent stakeholders to share and use data in a federated environment without centralising control over data assets. It translates the conceptual, semantic and technical foundations defined in WP3 into a coherent governance model that ensures coordinated, secure and compliant data exchange across the CRMs ecosystem. It also adopts a federated and unincorporated data space model, reflecting the diversity of actors involved in the CRMs domain and the need to preserve organisational autonomy, regulatory diversity and commercial sensitivities. Governance is conceived as an enabling mechanism rather than a central authority, defining the conditions under which data can be shared, accessed and reused while allowing participants to retain full control over their own data, systems and strategic decisions.

A key contribution of this deliverable is the definition of the governance structure, including roles, responsibilities and decision-making mechanisms within the MVDS. Key roles such as data owner, data provider, data operator, data space administrator, governance authority and service provider are clearly identified, providing accountability and consistency across the complete data lifecycle, from data creation and onboarding to access, use and monitoring.

The governance framework also establishes the policy and contractual foundations required for trusted data exchange. It defines how data space policies, contracts and technical enforcement mechanisms interact to regulate access, usage and sharing conditions. By linking governance rules to policies, the framework operationalises data sovereignty, FAIR principles, interoperability requirements and regulatory compliance.

In addition, the deliverable clarifies the relationship between governance, data management and technical implementation. Governance sets the normative and organisational constraints, while data management and platform components implement these constraints through metadata standards, quality procedures, identity management, access control and audit mechanisms.

The governance framework presented in this deliverable is preliminary and evolutionary. It is expected to be refined as the MVDS is implemented, data flows are operationalised and regulatory or technical requirements evolve. It also provides a

reference for future exploitation and sustainability planning, particularly in relation to the Business Plan developed in WP7.

Overall, this deliverable establishes a robust, flexible and European-aligned governance baseline for the CRMs Data Space, enabling the project to proceed with platform deployment, integration and long-term planning under a shared, transparent and sovereignty-preserving governance model.

1 Introduction

Among the several Specific SMART objectives defined for the CRMsDataSpace in the project proposal, **Objective 1** is “To develop a data space platform aligned with the principles governing European data spaces to ensure data exchange, integration, and analysis between CRM stakeholders and facilitate creating a Minimum Viable Data Space (MVDS)”.

The goal of **Work Package N°4** is the development of this MVDS, its specific objectives being as follows:

1. Develop the governance framework to ensure the platform is managed effectively, efficiently, and following the participants' business objectives.
2. Build the data space platform using state-of-the-art technologies and architectures.
3. Connect diverse data sources, systems, and tools into the platform to enable seamless data exchange and collaboration.
4. Maintain the platform's performance, managing data governance.
5. Troubleshoot technical problems, provide user training, and ensure compliance with data security and privacy regulations.
6. Host and maintain the data space platform to ensure its availability, security, and optimal performance over time.

Deliverable 4.1 describes the fundamental rules of the governance framework that will serve as the initial basis for the creation of the MVDS, as per specific objective N°1.

This deliverable is an outcome of **Task 4.1 – Governance system design and implementation**, within which SATEC is developing a governance framework beginning with a clear translation of the project's semantic, organisational and regulatory foundations into a coherent set of rules, processes and structures that ensure coordinated, secure, and compliant operation of the data space. The work must take as starting points the conceptual model, data specifications and quality procedures defined in WP3, since these provide the architecture and behavioural constraints that governance must formalise.

Based on this foundation, the governance framework defines the organisational roles, decision-making mechanisms and collaboration structures, while incorporating essential principles such as data sovereignty, FAIR compliance, interoperability and security. This ensures that governance reflects both the technical design of the platform and the real operational requirements of the partners who will manage and maintain it.

The governance framework outlines the policies and operational rules that regulate how data enters, circulates within and exits the data space, as well as the rights and responsibilities of all stakeholders. This includes establishing procedures for access

control, compliance monitoring, quality assurance, lifecycle management and alignment with European Data Space principles.

Because the governance system will directly influence later exploitation strategies, it also addresses long-term sustainability, the potential role of new organisational structures (such as a spin-off), and the need for transparent, auditable processes. By integrating normative, organisational, semantic and technical dimensions into a unified governance blueprint, **the framework will provide a foundation for the development of the Business Plan** in WP7.

SATEC will lead this task. **Subcontractor CTIC will cooperate** with SATEC based on their expertise in data space development and implementation.

Finally, it must be noted that the governance framework submitted at this stage is preliminary and may be subject to changes as the project progresses. As development advances and requirements evolve, adjustments may be necessary to ensure alignment with project needs, data availability and technical constraints. This framework should therefore be **considered a first version rather than a final specification**.

2 Data space governance and interoperability

2.1 Concept of governance & Application to federated data spaces

From a scholar perspective, governance refers to the systems, processes and structures through which an organization, community or state makes decisions, exercises authority and manages resources. It encompasses the mechanisms of accountability, participation, transparency and rule-making that guide collective action.

Applied to data management, governance is the framework of policies, processes, roles, standards and technologies that ensure an organization's data is accurate, consistent, secure, usable and aligned with business objectives. It defines how data is collected, stored, accessed, protected and used and establishes accountability for data-related decisions across the organization. DAMA-DMBOK, a globally recognized standard for data management, defines governance as *"the exercise of authority, control and shared decision-making over the management of data assets"*.

In the context of data spaces, governance refers to the set of rules, principles, roles and processes that coordinate how multiple participants (organizations, public entities, private companies) share, access and use data within a federated ecosystem. It ensures that data sharing is secure, fair, interoperable and aligned with common objectives, while maintaining data sovereignty.

Regarding federated data spaces, governance focuses on coordinating independent participants who retain full control over their own data, while common rules, roles and technical standards ensure that interactions across the ecosystem remain interoperable, secure and trustworthy. Thus, the main objective of governance is to provide a structured framework for participation, compliance and dispute resolution that enables collaboration and data exchange without compromising institutional autonomy or data sovereignty.

2.1.1 Governance in a federated CRMs Data Space

A governance framework for a CRMs Data Space must balance the need for coordinated data exchange with the autonomy and commercial sensitivities of the participating organizations. Because the ecosystem spans diverse actors (mining companies, refiners, manufacturers, recyclers, governmental agencies, certification bodies and research institutions), the framework must clearly define roles, responsibilities and interaction rules.

These roles typically include participants who contribute information about extraction, processing or supply chains; who use this information for planning, compliance or risk assessment; who offer technical capabilities such as identity management or semantic

services; and governance bodies, which oversee rule-setting, compliance mechanisms and dispute resolution. Clear delineation of such roles supports accountability and creates predictable pathways for participation.

Policies form the backbone of this governance framework by establishing the rules that ensure interoperable, secure and trustworthy and data exchange. These policies may cover specifications such as data quality requirements, access control rules, semantic standards, security measures and procedures for onboarding or certification. Because the CRMs ecosystem includes both public-interest objectives (e.g., improving supply chain transparency, supporting strategic resource planning) and commercial concerns (e.g., protecting proprietary information or competitive positions), policies must enable selective data sharing, confidentiality protection requirements and strict adherence to data sovereignty principles. The governance framework must therefore articulate which obligations apply universally across the data space and which remain optional or context-specific.

In a federated CRMs data space, the concept of contract refers to an agreement that governs a specific data-sharing transaction, by establishing the conditions under which data may be accessed and used (such as permitted usage, retention periods, transformation rights, redistribution limits or any other required obligations) and binding these rules to the data itself through usage-control mechanisms. By expressing policies in a standardized and interoperable format, contracts ensure that every data exchange instance is transparent, enforceable and consistent across participants' systems. They function as a dynamic layer that enables fine-grained, context-specific governance, allowing organizations to share data in a controlled, sovereignty-preserving manner without relinquishing authority over their assets.

The scope of the governance framework extends to the coordination of technical standards, trust services, accreditation schemes and dispute resolution processes that enable reliable collaboration across organizational boundaries. Yet its authority has clear limits: it does not centralize data, override national regulations, or impose direct control over participants' internal data management practices. Instead, it focuses on defining the conditions under which data can be shared, accessed and reused within the shared ecosystem. Participants retain full autonomy over their own systems, data and strategic decisions, with governance acting as an enabler rather than a controller.

These limitations are not weaknesses but foundational principles of a federated approach. By respecting data sovereignty, industrial confidentiality and regulatory diversity, the governance framework ensures that participation remains voluntary, trustworthy and scalable. It establishes the guardrails and incentives that allow a complex, multi-stakeholder ecosystem to function without hierarchical control. In doing so, governance becomes a catalyst for improved transparency and resilience in critical raw materials supply chains, while safeguarding the autonomy of every actor involved.

Finally, the EU is promoting the creation of Common European Data Spaces, conceived as sector-specific frameworks to enable trusted, secure and interoperable sharing of data among businesses, public bodies, and citizens. They set common rules, standards, and governance so data can flow across the EU while respecting privacy, competition, and ethical requirements.

Though there is not necessarily a single, rigidly defined data space per sector, the EU's vision is that each domain has one overarching Common European Data Space framework, so there can be multiple technical implementations, platforms or ecosystems that interoperate under shared rules and standards within each domain. The goal of the CRMDS project is to provide this framework for CRMs.

2.2 Data space governance components

Governance in a federated data space requires a governance framework that coordinates interactions balancing collective coordination with individual autonomy. To that end, governance can be conceived as the combination of three complementary components, each addressing distinct aspects of policy, process and infrastructure necessary for reliable and scalable data sharing:

- **Strategic** governance defines the overarching purpose, principles and rules that guide a federated data space. It sets the vision, scope and long-term objectives, establishes the value propositions for participants, and determines the policies that ensure trust, fairness and data sovereignty. At this level, governance bodies define membership criteria, high-level compliance expectations, and the decision-making structures through which the ecosystem evolves. Strategic governance ensures coherence and direction while maintaining the decentralization inherent in a federated model.
- **Operational** governance focuses on the processes, roles and procedures that enable the data space to function on a day-to-day basis. This includes participant onboarding, certification, monitoring, dispute resolution, and the coordination of cross-organizational workflows. Operational governance also defines how policies are translated into practical rules, how usage policies are enforced, and how changes (e.g., standard updates or onboarding of new actors) are managed. In a federated environment, operational governance ensures that interactions remain consistent and trustworthy without relying on centralized control.
- **Technical** governance covers the standards, protocols and trust services that ensure secure and interoperable data exchange across independent systems. It defines the technical building blocks (see subsection on data governance layers below) that participants must implement to interact within the federated ecosystem.

Technical governance ensures that the infrastructure operates reliably and that enforceable, machine-readable policies accompany each data-sharing instance. By harmonizing technology choices while allowing participants to retain their own systems and data, it provides the foundation for sovereignty-preserving interoperability.

3 Main functions of the data space governance framework

The data space governance framework establishes guidelines that guide the functioning of the ecosystem, ensuring that interactions between participants are carried out under criteria of reliability, consistency and compliance with the principles governing federated data spaces. These functions are aimed at creating an environment based on trust, control, quality and sustainability, which are essential elements for ensuring that data exchanges take place within a federated model that respects the sovereignty of each entity.

The value of the governance framework lies precisely in its ability to enable sovereignty, interoperability, effective governance, trust and decentralisation among participants, ensuring that each organisation retains control over its digital assets while participating in a shared environment.

In this context, the main functions of the framework are described below:

- **Trust.** Governance establishes the rules and mechanisms necessary to create an environment of mutual trust. This function is embodied in clear rules, transparency in interactions, verifiable compliance, and guarantees that all participants act under equivalent principles. Trust thus becomes the foundation that enables collaboration between independent entities.
- **Sovereignty & Control.** The framework ensures that each organisation maintains control over its data, precisely defining how it can be used, under what conditions and with what limitations. This control encompasses both access management and the application of usage policies. Digital sovereignty is a key principle that ensures that rights and responsibilities always remain in the hands of those who share the data.
- **Quality Assurance.** Governance incorporates criteria and procedures that ensure an adequate level of quality in the ecosystem's data and processes. This includes semantic consistency, integrity, adequate updates, availability, and operational reliability. Quality is a prerequisite for data to be used reliably and for participants to develop services with guarantees.
- **Sustainability & Continuity.** The framework defines the mechanisms that enable the long-term sustainability of the data space, from both an organisational and technical perspective. It includes the ability to adapt to new requirements, update standards, incorporate new participants, and ensure operational continuity. Sustainability ensures the viability of the ecosystem over time, even in the face of regulatory, technological, or market changes.

These functions articulate a governance framework that not only guarantees sovereignty and control, but also aligns with the principles of semantic, technical and organisational interoperability defined in the previous conceptual framework, ensuring that the data space can operate in a federated, consistent and scalable manner.

3.1 Relationship between data space governance and data management

Data space governance and data management are structurally interrelated, given that the former establishes the framework of rules, policies and regulatory principles that must be complied with in the ecosystem, while the latter ensures their effective application through operational procedures. Governance sets the requirements for sovereignty, interoperability, use control, trust and integrity that characterise federated data spaces according to the European Blueprint, while data management implements them through the implementation of harmonised data models, controlled vocabularies, structured metadata and consistency and integrity mechanisms that ensure that information retains its meaning, structure and traceability throughout its life cycle. This relationship allows the access, use, quality and security policies established by governance to be directly reflected in practices such as data validation, cataloguing, versioning, auditing and preservation, ensuring compliance with FAIR principles. Their alignment becomes especially relevant when the data space governance and the data management are considered alongside the European Interoperability Framework (EIF, European Commission 2017), which defines the legal, organisational, semantic, and technical conditions required for interoperability across public and private sector systems in Europe.

Together, governance and data management form a coherent framework that enables data to circulate between heterogeneous systems in a secure, reliable manner and under the control of the provider, ensuring a federated ecosystem capable of operating with common criteria and evolving in a sustainable way.

4 Data space building blocks

Taking the OpenDEI *Design Principles for Data Spaces* as a reference (Nagel L. and Lycklama D., 2021), four main categories of building blocks are identified: three technical blocks; Data Interoperability, Data Sovereignty & Trust, and Value Creation, each structured into three sections, and a Governance block, also divided into three components; Business agreements, Operational agreements, and Organizational agreements. The *Blueprint 2.0* (Data Spaces Support Centre, DSSC 2025) proposes a similar architecture, but further compartmentalizes these blocks into smaller sections, enabling a customizable governance framework and technical foundation tailored to the specific use case of each data space.

In line with this approach, this document recommends adopting the DSSC model, as shown in the figure below, adapting each block to the proposed use case for developing the governance of a Minimum Viable Data Space (MVDS). These layers reflect established practices in European data space initiatives and provide a consistent framework to facilitate the secure exchange of data and the management of responsibilities.

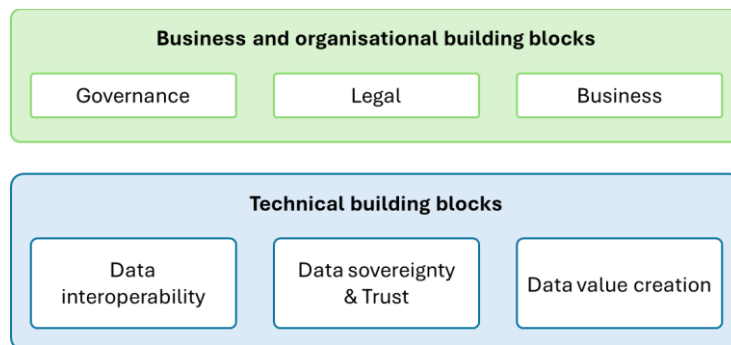


Figure 4-1. Data space building blocks (Modified after DSSC, 2025)

The following sections of this document address the business and organisational building blocks, with particular emphasis on the governance block, detailing the essential elements required to establish a MVDS. They will elaborate on the functions of the governance authority, the roles and responsibilities of the project, and the participation management process.

The technical building blocks will also be covered, albeit more briefly, given their specific focus on interoperability, sovereignty, and identity; key capabilities for ensuring the correct and trustworthy operation of a data space.

It is important to highlight that the building blocks are strongly interconnected, both within the same category (e.g., governance with business) and across different categories (e.g., governance with identity and trust).

4.1 Business and organisational building blocks

4.1.1 Governance building block

This block establishes the governance structure, roles, and organisational processes that govern the operation and evolution of the data space. It ensures transparency and consistent management throughout the ecosystem's lifecycle, facilitating decision-making and the incorporation of new participants.

The governance framework proposed for the CRMs data space adopts the model of an unincorporated data space. In this approach, cooperation structures such as unincorporated joint ventures, consortia, or similar arrangements are entirely member-driven, which makes robust governance mechanisms essential to align and coordinate the activities of all participants. These mechanisms focus solely on achieving the shared objectives of the cooperation while avoiding any management of members' resources, which remain under their full control, or any governance over results, whose ownership also stays with the members.

The **governance authority** defines the **governance model and responsible bodies**, i.e., committees, working groups or entities responsible for coordinating the technical, legal and strategic aspects of the data space are defined. Each body must have a defined mandate, voting mechanisms and defined responsibilities. It also establishes the **different roles and associated responsibilities** within the ecosystem, describing their scope of action, level of participation and degree of authority. Likewise, the competences assigned to each role and the obligations they must fulfil within the data space governance framework are detailed.

4.1.1.1 Organisational form and governance authority.

Data spaces can be established as incorporated data spaces (with a legal entity) when they are designed to operate permanently and with a clear business orientation. Alternatively, they may begin as unincorporated data spaces (without a legal entity), serving as an initial collaborative framework that may later evolve into an incorporated form. In this second model, governance relies heavily on agreements between the data space participants. For instance, some European Union Member States allow the creation of joint ventures that do not constitute separate legal entities, where participants simply share resources and knowledge. Other forms of cooperation or collaboration agreements, such as strategic alliance agreements or consortium

agreements, are also possible. For simplicity, all these arrangements are referred to as founding agreements, and their signatories as data space members. By signing the founding agreements, members assume specific obligations toward one another in pursuit of the shared goals of the data space.

The following table presents a comparative overview of the differences between these two approaches (DSSC, 2025).

Table 4-1. Comparison of unincorporated and incorporated data spaces

Aspect	Unincorporated data space	Incorporated data space
Temporal existence	Data space is a temporary establishment, although it may exist for several years. The termination is pre-determined by the founding agreement.	Data space is a permanent establishment; its termination is not pre-determined by the founding agreement.
Legal personality	Data space is not a legal person.	Data space is a legal person, separate from its members.
Ability to enter into contracts	Data space cannot conclude contracts with third parties in its own name. Only members can conclude contracts.	Data space can conclude contracts in its own name with third parties.
Ownership of property	All property is owned and disposed of by members, individually or jointly.	Data space owns (intellectual) property and disposes of it.
Liability	Members are liable for their actions. Data space cannot bear any liability.	Data space bears liability for the actions of its governance bodies.
Complexity of the founding agreement	The founding agreement is a complex document regulating various aspects of relationships between members.	The founding agreement is fairly simple document establishing the data space and regulating only certain monetary relationships between the data space and its members.
Control over the data space	Members are in complete control of the data space.	Data space is in control of itself through its governance bodies.
Addition of new members	The process requires consent of all existing members and re-negotiation of the founding agreement.	The process varies depending on the legal form but usually does not require re-negotiation of the founding agreement.

Regardless of the chosen model, the establishment of a governance authority within a data space is strongly guided by internal rules such as founding agreements, contractual frameworks for onboarding participants, internal policies, governance procedures, and technical specifications, as well as by external regulatory frameworks governing areas such as intellectual property, digital rights, personal data protection, platform regulation, trust and security, or sector-specific rules. Within the boundaries set by the founding agreement and applicable law, each data space can and should define its own terms and conditions of use, internal regulations, and operational policies, along with diverse governance procedures (e.g., dispute resolution, updates to technical specifications, and others). All these elements must be included in the agreement that every participant is required to sign prior to joining the data space and engaging in data transactions.

Within an unincorporated data space, participants are free to establish a governance authority tailored to their needs and preferences, as no legal requirements dictate equal or specific forms of representation in decision-making. Ultimately, founding agreements in such a model are, though complex in content, relatively easy to amend, allowing the data space to adapt flexibly to evolving project circumstances and needs.

In the context of the CRMDS project, SATEC acts as the governance authority; it will be the entity responsible for promoting changes in governance based on agreements between data space participants.

4.1.1.2 *Minimum Viable CRMs Data Space roles and responsibilities.*

Participants in the CRMs Data Space each play essential, interconnected roles that ensure the ecosystem functions smoothly and responsibly. Their collaboration enables trusted data sharing, supports interoperability, and maintains the balance between openness and control. By fulfilling their respective responsibilities, they all contribute to create a secure, reliable and value-generating environment where data can flow confidently across organizational and sector boundaries.

- **Main roles within MVDS data workflow.** The governance framework defines a set of roles as shown in the following figure. Roles are shown in **bold** for ease of reference.

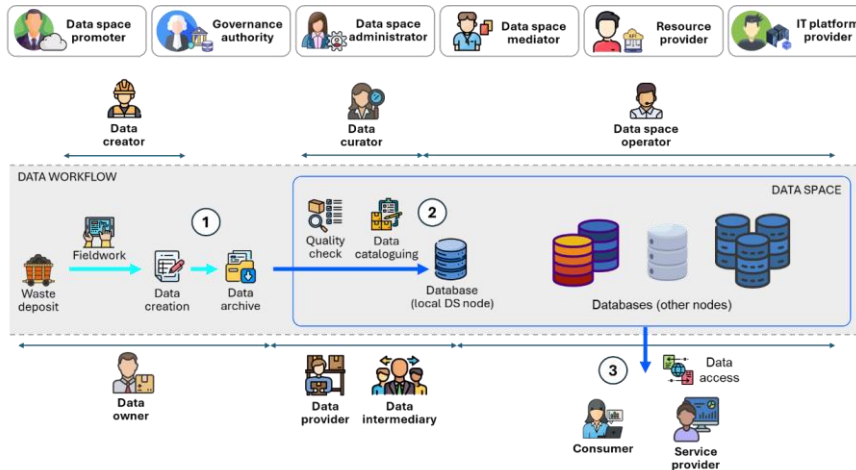


Figure 4-2. Roles involved in the Minimum Viable CRMs Data Space

The data workflow is as follows:

1. **Data creation:** data are generated through fieldwork conducted at the waste deposit. This work may be carried out directly by the **data owner** or by a designated **data creator** contracted for that purpose. Once produced, the data are stored in the data owner's archive.
2. **Data onboarding to the data space:** The data are then submitted for inclusion in the data space catalogue either by a **data provider** (the data owner, if they operate their own data-space node) or by a **data intermediary** (the operator of a node made available for the owner's data). Before onboarding, the data undergo a quality check to ensure compliance with data space requirements. If approved, they are added to the catalogue and become officially discoverable by participants. Both the quality check and the cataloguing are handled by a **data operator**.
3. **Data sharing via the data space:** consumers (agents wishing to inspect the data) and **service providers** (agents using the data as inputs to produce new outputs) request and access data through the data space. The data-sharing environment is monitored and maintained by a **data space operator**.

A set of additional roles is required to ensure the data space is effectively implemented, managed and governed: **data space promoter**, **governance authority**, **data space administrator**, **data space mediator**, **resource provider** and **IT platform provider**.

MVDS roles and responsibilities. The table below summarises all roles defined within the CRMsDS, highlighting their key characteristics and illustrating each with concrete examples from the CRMsDS project. It serves as a reference for understanding how responsibilities are distributed across participants and how each role contributes to the effective operation and governance of the data space. As with other aspects of governance, this proposed set of roles reflects the current stage of the project but may be revised as the project progresses.

Comentado [C01]: Propuesta CTIC

Table 4-2. Roles within the CRMs Data Space

Role	Exclusive / Shared	Description	Example
Consumer	Shared	Requests data access for data inspection or analysis	GIG-PIB INSEMEX THGA UBU UNIOVI
Data creator	Shared	Performs fieldwork or other activities that generate new data	
Data intermediary	Shared	Shares third-party data through its own data space node	
Data operator	Exclusive	Verifies that data comply with data space governance rules before cataloguing them	SATEC
Data owner	Shared	Holds legal ownership of the data	GIG-PIB INSEMEX THGA UBU UNIOVI
Data provider	Shared	Shares its own data through its own data space node	
Data space administrator	Exclusive	Oversees the day-to-day operation of the data space	SATEC
Data space mediator	Exclusive	Manages conflict resolution between data-space participants	SATEC
Data space promoter	Exclusive	Initiates and steers the data space project	SATEC
Governance authority	Exclusive	Holds the mandate to enact changes to governance, according to prior agreements among participants	SATEC
IT platform provider	Exclusive	Develops and maintains the IT infrastructure (hardware, software, and communication layers) supporting the data space operations	SATEC
Resource provider	Shared	Supplies specialised components required for specific uses (e.g., APIs enabling integration with advanced tools such as AI)	UBU
Service provider	Shared	Uses the data as inputs to produce new outputs (data, services, insights, etc.)	UBU

4.1.1.3 Participation Management

The Participation Management block outlines the governance processes for managing how participants engage with a data space. It provides guidance on enabling, tracking, organising, and ultimately managing participant involvement, from onboarding and integration to secure interaction and offboarding, ensuring smooth operational and governance outcomes. Participants include organisations and individuals who engage in data transactions, provide services, or hold data rights. Successful participation requires a shared mission or purpose within the data space and can deliver value from data or compelling incentives to join. Efficient onboarding, ongoing organisation, and defined mechanisms for exit are all critical parts of enabling collaboration.

The elements of the participation management are (DSSC, 2025):

- **Participants & Roles:** Identifies different participant types such as data providers, data users, intermediaries/operators, data rights holders, and possibly external stakeholders whose interests impact the ecosystem. Role differentiation (e.g., governance authority members vs. other participants) helps clarify responsibilities and maintain trust.
- **Onboarding:** Is a critical process that ensures participants can join a data space smoothly while meeting all necessary legal, organisational, and technical requirements. The data space governance authority defines the minimum participation criteria, primarily through the General Terms and Conditions (GTCs), which specify roles, responsibilities, admission policies, identity verification rules, attestation requirements, technical standards, data protection obligations (including General Data Protection Regulation (GDPR, EU 2016) compliance where relevant), and offboarding regulations. These terms reflect the data space's purpose and regulatory context, balancing low entry barriers with the need for interoperability, data quality, and security.
 - ✓ The onboarding procedure is governed by **pre-conditions**, including verifying eligibility, identity, compliance, and technical readiness, as well as acceptance of the GTCs, either through a formal application process or automated acceptance. The level of verification depends on the participant's intended role. Once accepted, participants receive Verifiable Credentials (VC), and relevant information is published in the data space registry. To illustrate some real cases within the CRMDS, a mining institute, for example, must demonstrate its ability to share datasets using the standard formats defined in the data model (INSPIRE, GeoSciML, among others), as well as comply with trust rules and adhere to established standards (Gaia-X Trust Framework, eIDAS, among others). In the case of a research centre, it must demonstrate security readiness

for handling potentially sensitive data that is either being considered for research or already under active study.

Comentado [C02]: Propuesta CTIC

- ✓ **Post-conditions** ensure full integration, including technical onboarding support to guarantee interoperability, data quality, and correct service configuration. Continuous monitoring is essential to confirm ongoing adherence to policies and standards. Participant feedback helps refine the onboarding process over time, ensuring that it remains efficient, secure, and aligned with evolving needs and regulatory changes. Within the scope of this project, a mining institute must also prove its ability to integrate metadata into the data space catalogue and ensure subsequent interoperability across endpoints. A research centre must demonstrate its integration with the data consumption APIs and its adherence to the licence terms defined for research use and data publication.

Comentado [C03]: Propuesta CTIC

- **Data transaction:** Beyond onboarding, this element includes enabling participants to engage in data transactions while upholding governance, data quality, interoperability, and legal compliance. As the data space grows, the data space governance authority needs to regularly screen the governance framework and eventually adapt it to address emerging needs and challenges.
- **Offboarding:** Outlines processes for participants exiting the data space, including handling remaining obligations, data rights, and maintaining ecosystem integrity during transition. More specifically, the offboarding process should consider:
 - ✓ **Documentation of exit procedures:** A well-defined and documented offboarding procedure ensures consistency and completeness. It outlines the required steps for data transfer, access termination, and contract closure. This includes continuous lifecycle management of credentials, as defined in the Identity and Attestation building block.
 - ✓ **Data transfer and deletion protocols:** Clear protocols must guide the secure transfer or deletion of data when a participant exits. Data should be transferred to another party or deleted according to the agreed licences, data space policies, and applicable legal requirements. This ensures data sovereignty and regulatory compliance.
 - ✓ **Notification:** All relevant stakeholders must be promptly notified of the participant's exit. Notifications should include information about data handling, access termination, and any outstanding obligations to prevent misunderstandings and maintain operational continuity.

- ✓ **Verification of compliance:** A final compliance check confirms that all contractual, financial, and legal obligations have been fulfilled before completing the offboarding. This includes ensuring that all agreements with other participants have been properly closed.
- ✓ **Offboarding support:** Support should be provided during the transition to address issues or questions, such as assistance with data transfer or guidance on remaining obligations. This helps ensure a smooth and cooperative exit process.
- ✓ **Periodic framework reviews:** Regular reviews of the data space's governance framework help incorporate updates due to legislative or operational changes. These reviews ensure that offboarding and onboarding processes remain sustainable, compliant, and effective over time.
- **Continuous Monitoring:** Participation should be tracked, reviewed, and updated as needed to ensure ongoing relevance and compliance with rules and regulations

4.1.1.4 Relationships with other building blocks

To understand how the data space's building blocks interact with each other it is necessary to identify the specific sub-blocks that make up each of the six categories shown in the next figure. These blocks are listed in the figure below but not developed in full in this document. Each of the sub-blocks referenced in the next figure will be detailed, if necessary, as the data space governance framework matures, ensuring that it is supported by a complete, robust, and interconnected set of organisational, technical, and legal functions.

The **Organisational form** building block is closely interconnected with several other foundational elements of the data space. Its design is strongly influenced by the *business model*, as decisions made by the data space members, such as whether the initiative is temporary or permanent, where it will operate, and whether it is profit-oriented, directly shape the choice of organisational and legal form. Conversely, constraints imposed by applicable legal frameworks may require adapting the business model. Governance also underpins *participation management*, since the selected organisational form brings specific legal obligations that guide how new members are admitted, how they participate in the governance authority, and what roles, rights, and duties they hold within the data space.

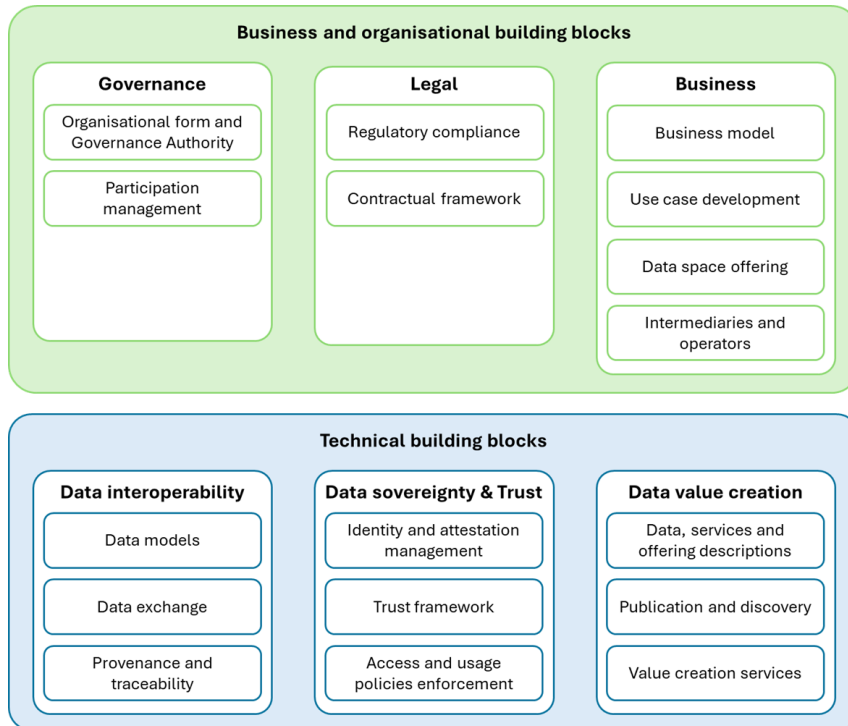


Figure 4-3. Data space building blocks and sub-blocks (Modified after DSSC, 2025)

Similarly, the *contractual framework* depends on the governance structure: contracts define the rights and obligations of members and their relationship with the data space, but the nature of these contracts, including contracting parties and liability arrangements, is determined by the organisational form established in the governance building block. In this way, governance acts as the structural anchor connecting business decisions, participation rules, and contractual relationships across the data space.

Participation management it is also closely interconnected with several other data space building blocks, each contributing essential components for onboarding, operation, and offboarding of participants. The *access & usage policies enforcement* building block provides the technical mechanisms required for applying access and usage rules, which are fundamental during both onboarding and offboarding. The *contractual framework* supports participation management by defining the legal instruments, such as GTCs, access agreements, and data transaction agreements, that formalize participant commitments and enforce the processes defined in participation management. The *organisational form and governance authority* building block is

equally central, as it outlines the internal rules, responsibilities, and governance structures established by the data space Governance authority, directly shaping participation requirements.

Participation Management also depends on strong *regulatory compliance*, ensuring that both the data space and its participants adhere to applicable laws, including data protection, privacy, and sector-specific regulations. The *identity and attestation management* building block underpins trustworthy participation by requiring secure identification, credential lifecycle management, and attestations aligned with eIDAS (EU Directive 2022/2555) and trusted service provider lists. These attestations confirm adherence to the governance framework and are recorded in the data space registry.

Furthermore, *intermediaries and operators* play a functional role in participation, as they provide services that support data transactions and operational processes under the governance framework. Finally, *data provenance and data traceability* extend beyond technical tools to include governance processes that ensure regulatory and contractual compliance throughout participation.

4.1.2 Legal building block

The legal building block supports a clear understanding of legal and contractual requirements within a data space. More specifically, the **regulatory compliance** block identifies the key elements needed to ensure adherence to legal obligations and focuses on legal triggers that indicate when specific regulatory frameworks apply. The **contractual framework** block provides an overview of essential agreements highlighting common cross-sector issues such as fairness and clarity.

Together, these said blocks offer foundational guidance for organizing compliance and contractual structures across different types of data spaces, without prescribing exhaustive templates.

4.1.2.1 Regulatory compliance

This building block provides guidance to the data space governance authority on how to apply legal requirements to the design and operation of a data space. It supports the proper definition of participant roles and responsibilities, the creation of internal policies, and the continuous monitoring of regulatory compliance. It also helps participants understand their rights and obligations under relevant regulatory frameworks and informs prospective participants, developers, and policymakers about applicable legal considerations.

Among other mechanisms, this block defines *triggers* as events that occur in a specific context in a data space and marks that a particular legal framework could be applied.

The regulatory compliance building block covers all activities needed to ensure that a data space operates in accordance with applicable legal frameworks. This involves identifying relevant regulations, understanding how they affect the data space's functions and participants, and ensuring ongoing compliance throughout the data space lifecycle. The governance authority is responsible for determining which legal rules apply to its specific data space. The next figure illustrates how multiple legal frameworks may apply simultaneously to a data space without a tie to a specific sector.

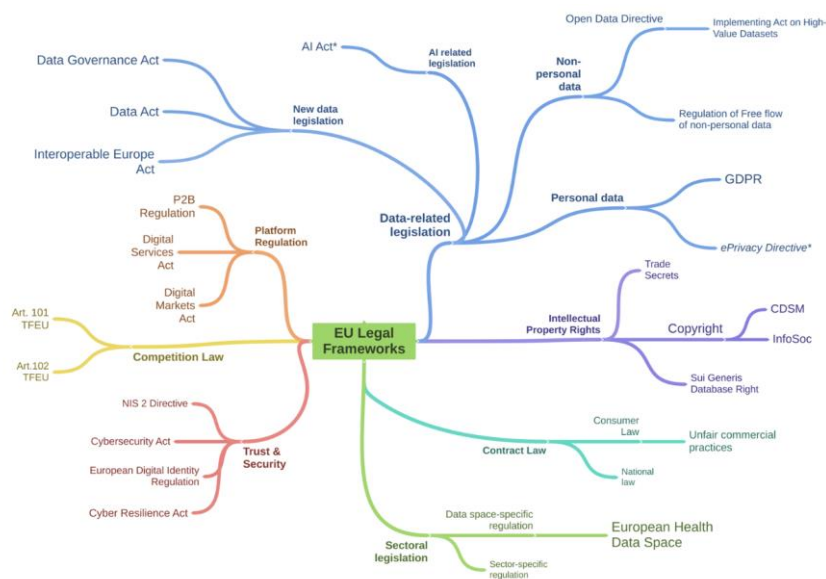


Figure 4-4. EU Legal frameworks (DSSC, 2025)

In the context of this project, several regulatory frameworks are relevant, including the Critical Raw Materials Act (CRMA), the United Nations Framework Classification for Resources (UNFC), and technical regulations such as INSPIRE (EU, 2007) and open data standards, among others. A clear example of cross-cutting applicability is the AI Act (EU, 2024), which becomes particularly important due to its connection with services that rely on data exploration and exploitation capabilities in an environment governed by sovereignty and trust.

4.1.2.2 Contractual framework

The contractual framework defines all agreements that regulate how a data space operates and how its participants interact. It creates legally enforceable relationships, rights, duties, and governance mechanisms among participants, members, and enabling

service providers. Beyond implementing the governance framework, it also allows participants, particularly data providers, to exercise data sovereignty by adding conditions to the baseline governance set by the data space authority. The framework is broad and is therefore organised into three main categories:

- **Institutional Agreements.** These agreements establish the data space and set the mandatory governance rules applicable to all participants. They define the common contractual foundation, regulate relationships among members, and may prescribe standardised terms, licences, or clauses to be used in data product contracts. By introducing shared contractual elements across the data space, institutional agreements reduce legal complexity, improve interoperability, lower transaction costs, and ensure regulatory compliance. They translate organisational and business decisions into uniform legal obligations.

- a. **Founding agreement.** The founding agreement is the instrument that formally establishes the data space and its governance authority. It is signed by the founding members and defines the purpose and objectives for which the data space is created. Depending on the chosen organisational form, the agreement may take different legal shapes.

The agreement typically includes clauses describing the background and purpose of the data space, identification of the parties involved, general responsibilities and liabilities, and references to the governance model chosen. It also defines the admission policy for members, identifying the initial signatories who will form the governing body and, where applicable, specifying conditions for new members to join in the future. Implementation of these membership rules depends on the legal form chosen.

- b. **Terms and conditions.** The General Terms and Conditions (GTCs) form an agreement that gives legal enforceability to key elements of the data space governance framework, making certain rules and processes binding on all participants. They may be incorporated into the founding agreement or referenced by it. The GTCs regulate interactions at the data space level, defining the roles and responsibilities of all actors, governance authority, data providers, data consumers, and service providers.

At the transaction level, the GTCs provide a common legal foundation for data transactions by introducing harmonised elements such as standardised clauses or licence typologies. This improves legal interoperability, reduces transaction costs and contractual complexity, and lowers the likelihood of disputes.

Key areas regulated at the data space level include admission policies, intellectual property rules, data protection policies (including GDPR), technical standards, cybersecurity and risk management requirements, and complaints and dispute resolution procedures. At the transaction level, the GTCs may coordinate or constrain the terms used in data-sharing and service agreements, for example by mandating specific licence sets or prohibiting certain contractual limitations. Agreement-specific clauses typically distinguish between rules applicable at the data space level and those applying to individual data transactions.

- c. **Agreement related to enabling services.** These are established by the data space governance authority to ensure the provision of essential operational services. These agreements typically cover three categories: data-related services, trust framework services, and identity management services.
- **Data sharing agreements.** These agreements govern the actual sharing of data, whether data is supplied, transferred, ported, or otherwise accessed, between data transaction participants. They do not bind the entire data space but only the parties involved in a specific transaction. Typically, the data provider defines the terms, reflecting data sovereignty, although the data space may also offer standard clause sets or templates. A specific form of these agreements is the data product contract, which contains the terms and conditions attached to a data product, including licensing, metadata, usage rules, and delivery mechanisms.
- **Services agreements.** Service agreements cover services provided within the data space. They include:
 - a. Data-related services, such as data processing, data trust/escrow, and data intermediation (e.g., data marketplaces, Personal Information Management Systems-PIMS, data cooperatives).
 - b. Enabling services, which support the technical functioning of the data space, such as federation services, participant agent services, and value-creation services.

4.1.2.3 Policy basics and lifecycle

The policy lifecycle begins with the **policy definition**, where the data space governance authority and participants collaboratively establish the rules governing data access, usage, security, and compliance. This stage involves identifying policy objectives, expressing them in machine-readable formats (such as XACML or ODRL/JSON-LD), and aligning them with legal, organisational, semantic, and technical requirements.

Once defined, policies progress to **application**, where they are enforced during actual data exchanges. Policy engines, connectors, and negotiation protocols evaluate whether a data transaction complies with the defined rules. This ensures that only authorised participants, roles, and purposes are permitted and that contractual or usage conditions are respected throughout the data exchange lifecycle.

The next stage is **audit**, which involves systematically recording and monitoring policy enforcement actions. Logs generated during negotiation, contract agreement, and data transfer are analysed to verify compliance, detect anomalies, and ensure that obligations, restrictions, and purposes were properly applied. This stage provides transparency, traceability, and accountability across the data space.

Finally, policies enter the **review** phase, where governance bodies assess whether existing rules remain valid, effective, and aligned with evolving legal, technical, or organisational needs. Feedback from audits, incidents, and participant experience informs policy updates. This ensures that the governance framework adapts to new risks, new services, technological innovations, and regulatory changes.

4.1.2.4 Policy types

The following sections detail the types of policies applied within a data space, covering both the legal and technical dimensions.

- **Data access and usage policies** regulate how data may be accessed, reused, and distributed within the data space. They define licensing conditions, rights of reuse, and permissible purposes, ensuring that data users understand the scope and limitations associated with each dataset or service. These policies can be expressed in machine-readable form using Open Digital Rights Language (ODRL), allowing licences, permissions, prohibitions, and duties to be formally encoded and automatically enforced. Together, these rules form the legal and contractual basis for compliant data consumption.
- **Data sharing and sovereignty policies** establish the principles governing data custodianship, ownership, and control. They enable data providers to define how their data can be shared, under what terms, and with which obligations, ensuring the protection of sovereignty rights. These policies also specify mechanisms such as consent, usage restrictions, and revocation conditions.
- **Data quality and validation policies** define requirements for accuracy, consistency, integrity, and traceability of the datasets exchanged within the data space. They describe validation rules, quality metrics, metadata completeness criteria, and provenance requirements to ensure that data consumers receive

reliable, trustworthy data while enabling auditing and quality assurance processes.

- **Security and privacy policies** ensure compliance with the General Data Protection Regulation (GDPR), confidentiality rules, and broader cybersecurity frameworks. They specify how personal and sensitive data should be processed, protected, and anonymised; how access must be controlled; and how security risks must be mitigated. These policies also define incident management, encryption requirements, and responsibilities for data controllers and processors.
- **Interoperability and standards policies** establish the technical and semantic frameworks required to enable consistent cross-domain and cross-border data exchange. They define mandatory standards, such as INSPIRE for geospatial data, DCAT-AP (EU, 2025) for data catalogues, and the Open Geospatial Consortium (OGC) standards for spatial services, to ensure compatibility, semantic alignment, and reusable data structures across the ecosystem.

The policies in the CRMDS can be categorised according to their level of necessity. Three categories can be established, always beginning policy development with those that are mandatory, while allowing both their scope and nature to evolve as the project progresses.

- **Mandatory policies:** those related to security, identity, access control for datasets, usage licences, and similar foundational requirements.
- **Recommended policies:** for example, those concerning data quality, lineage, and traceability, always establishing minimum requirements that a trusted data space participant must meet and that will be evaluated during onboarding.
- **Advanced policies:** for example, those relating to monetisation and the business model.

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4.1.2.5 Data space policies and product contracts

In a data space data transfer process, the data space policies, contracts, contract negotiations and contract agreements form the core mechanisms that regulate how data is exchanged and used.

- **Data space policies** define the overarching rules that govern how data may be accessed, exchanged, and used within the data space, establishing the baseline conditions applicable to all participants.

- **Contracts** determine which specific policies apply to each data asset or resource, translating general governance rules into concrete terms for individual datasets, services, or interactions.
- **Contract negotiations** constitute the process through which connectors exchange information, validate compliance with the relevant policies and contractual conditions, and ensure that all requirements are satisfied prior to initiating a data transfer.
- **Contract agreements** represent the final approval of the data exchange; they formalize the accepted terms and become the binding record of the transaction within the data space.

Together, these elements act as coordinated logs documenting every stage of the data transfer process, thereby ensuring auditability, traceability, and compliance. The following figure summarizes the policies workflow applied to a specific asset in a data space.

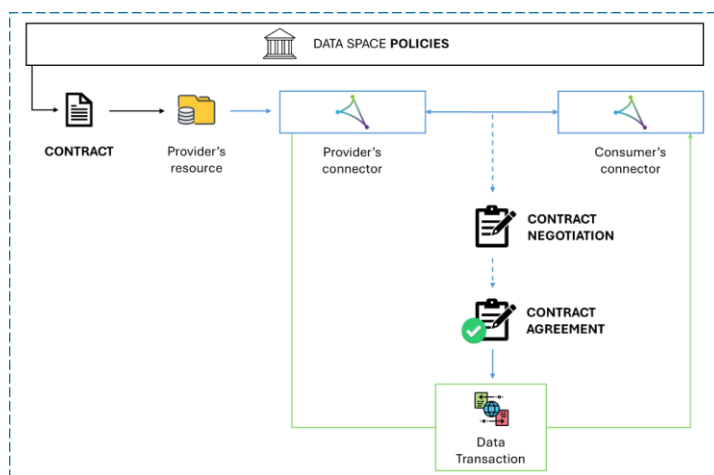


Figure 4-5. Policies and contracts involved in a Minimum Viable CRMs Data Space

4.1.3 Business building block

The purpose of this building block is to support strategic decision-making. Regardless of whether the orientation is public-sector or private-sector driven, the proposals within this block define the framework conditions that govern participation and commercial relationships in the data space. These building blocks link the data space's products and

services with operators and intermediaries through specific use cases. Although all business-related topics and more specifically, the business model, fall outside the scope of this document and will be defined in detail at later stages of the project, more information of the future alignment with the governance model is discussed in section 6.2.1 of this text.

In any case, a brief description of each of the building blocks included in this section is provided below.

- **Business Model.** This building block provides the foundation for ensuring the long-term sustainability of a data space by defining how value is created, delivered, and captured across a multi-actor ecosystem. Unlike traditional business models, a data space must balance economic viability with governance, interoperability, and trust while facilitating collaboration among data providers, users, and service intermediaries. Together, these organisations aim to create platform-like dynamics that connect the supply and demand of data and related services, while preventing any single actor from holding disproportionate power or influence. This approach is referred to as a collaborative, multi-sided business model. Although interrelated, the data space business model is distinct from the business model of individual data space participants, as shown on the next table.

Table 4-3. Business models in data spaces (DSSC, 2025)

Business Model Type	Scope	Key Focus Areas
Data Space Business Model	Entire data space ecosystem	Value creation for use cases, governance, revenue models, participant incentives
Participant Business Models	Individual organizations within the data space	Data monetization, service offerings, competitive differentiation

- **Use case development.** Use cases are specific settings in which two or more participants create business, societal, or environmental value from sharing data, and they are essential for attracting participants and enabling growth, especially in early phases. The process includes identifying and monitoring use case scenarios, refining these scenarios into detailed use cases by clarifying purpose, value, participants, and data flows, and then implementing the most viable use cases. A key role in this process is the use case orchestrator, who coordinates relevant actors and ensures that business rules and conditions are established and followed. This building block also emphasises finding synergies between use

cases, supporting value sharing among participants, and ensuring alignment with governance principles and rules at both the data space and use case levels. Overall, the use case development building block helps operationalise data sharing scenarios, connect them with functional requirements, and strengthen the data space ecosystem.

- **Data space offering.** Offerings contain data products, services, and the related descriptions that provide all the information needed for a potential consumer to decide whether to consume the data products and/or services or not. Data products are assets that deliver value by packaging data and metadata (including licence terms, quality, format, rights, delivery options, etc.) in a consumable form. Services are functional capabilities offered to participants to support data usage, insights, analytics, or other operational needs. Together, these products and services form the data space offering, which is curated to align with use cases and to foster network effects between data providers and consumers. The governance authority is encouraged to establish rules for onboarding priority data products and services, enforce governance of the offerings, and support participants in developing high-quality data products and services to be included in the offering. Thus, the data space offering connects the business needs of participants with the technical and governance structures required to make data products and services discoverable, usable, and valuable within the data space ecosystem.

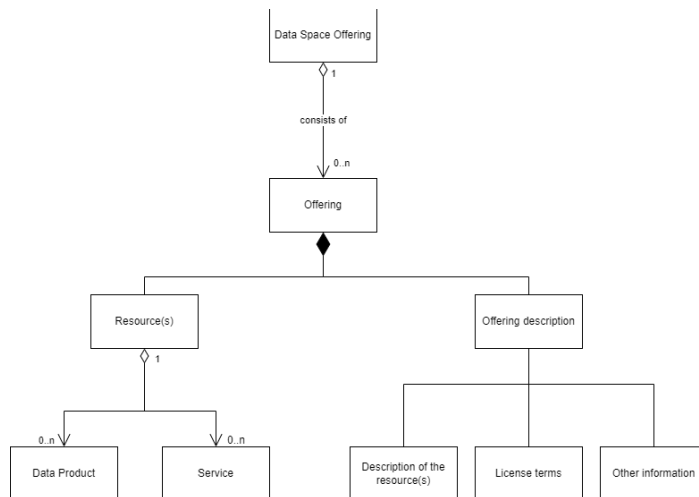


Figure 4-6. Data space offering, data offerings and related definitions (DSSC, 2025)

- **Intermediaries and operators.** A key design decision for any data space is how to leverage service providers to supply essential technical, business, and organisational services. While the data space governance authority and participants may provide these services themselves, outsourcing them can offer significant benefits. Value lies in simplifying trusted data sharing, enhancing accessibility and usability, supporting scalability, and enabling interoperability across data spaces. However, risks such as vendor lock-in, added vendor management costs, and potential loss of sovereignty must be taken into account.

4.2 Technical building blocks

Although they are not the focus of this document, the technical building blocks are strongly connected to the business and organisational building blocks and to the governance model specifically. For this reason, a brief description of key aspects of the interoperability, sovereignty and trust, and data value creation building blocks will be provided.

4.2.1 Data interoperability building block

Interoperability is composed of four layers: technical, semantic, legal, and organisational. The first two, technical and semantic interoperability, are addressed earlier in document *D3.2 Data space contents*. This chapter introduces the remaining two layers, legal and organisational interoperability, which are essential for enabling effective collaboration, compliance, and coordination among participants within the data space.

In addition to the four layers of interoperability, there is a cross-cutting component known as *integrated public service governance*, which spans and influences all four layers. Complementing these is a background layer, *interoperability governance*, which provides the overarching coordination, policies, and management structures needed to ensure that interoperability is maintained effectively across the entire system, as shown in the figure below.



Figure 4-7. Layers of interoperability (EIF, 2017)

Interoperability governance encompasses the decisions, frameworks, organisational structures, roles, responsibilities, policies, and agreements needed to ensure and monitor interoperability at national and EU levels. It is supported by key EU instruments such as the European Interoperability Framework (EIF), the Interoperability Action Plan, and the European Interoperability Reference Architecture (EIRA, 2024). Domain-specific frameworks like the INSPIRE Directive also demonstrate how legal, organisational, and technical interoperability can be coordinated effectively. Achieving interoperability requires strong political support, shared vision, aligned priorities, and sufficient resources across administrative levels. A lack of in-house expertise is a common barrier, making it essential for Member States to include interoperability skills in their national strategies. Because many European public services rely on shared components, their long-term sustainability must be ensured through durable interoperability agreements and coordinated governance. Interoperability governance provides the holistic approach needed, integrating coordination, communication, and monitoring. The European Commission supports this through initiatives such as the National Interoperability Framework Observatory (NIFO), which helps Member States align national frameworks with the EIF and share best practices.

Integrated public service governance ensures that multiple public administrations can work together to deliver seamless, user-centred European public services. When several organisations are involved, coordinated governance is required to plan, implement, and operate services in an integrated manner, ensuring reuse of data and services, interoperability, and the development of shared building blocks. Governance must cover all interoperability layers and must be continuously maintained, as interoperability can be disrupted by changes in legislation, administrative structures, business needs, processes, or technology. Effective integrated governance involves establishing organisational structures, roles, responsibilities, and decision-making processes; setting interoperability requirements for quality, scalability, and availability; defining service-level agreements for internally reused and externally sourced services; and managing

interoperability agreements and change processes. It must also include business continuity and disaster recovery plans to ensure that digital public services and core building blocks remain operational under adverse conditions, such as cyberattacks or system failures.

4.2.1.1 Legal interoperability

Legal interoperability ensures that public administrations, each operating under their own national legal frameworks, can work together to deliver European cross-border public services. It requires that legislation does not hinder cooperation between Member States and that clear agreements exist for handling legal differences, including the possibility of creating new laws when needed. The first step is to perform *interoperability checks* to detect legal barriers such as data-use restrictions, inconsistent licensing models, outdated data protection requirements, or regulations mandating specific technologies. Legislative coherence must be assessed both before adoption and throughout implementation. Because European public services rely heavily on digital delivery, proposed laws should undergo a *digital check* to confirm they fit digital contexts, identify obstacles to digital exchange, and evaluate Information and Communication Technology (ICT) impacts. This approach strengthens interoperability at semantic and technical levels and promotes reuse of existing ICT solutions, reducing cost and implementation time. Finally, the legal validity of exchanged information must be preserved across borders, and data protection rules must be respected in both sending and receiving countries, which may require supplementary agreements to address differences in national implementation.

4.2.1.2 Organisational interoperability

Organisational interoperability concerns how public administrations align their business processes, responsibilities, and expectations to achieve shared, mutually beneficial goals. It requires documenting, integrating, and harmonising business processes and the information exchanged between organisations, while ensuring that services are user-focused, accessible, and easy to identify.

- **Business process alignment.** It involves agreeing on common modelling techniques and documenting processes and information flows so all administrations involved can understand the complete end-to-end service and their respective roles.
- **Organisational relationships** between service providers and service consumers, formalised through mechanisms such as Memoranda of Understanding (MoUs) and Service Level Agreements (SLAs).

For cross-border services, these relationships are best supported through multilateral or EU-wide agreements.

4.2.2 Data sovereignty and trust building block

The contents of this building block ensure the reliability and quality of participant information within the data space, while simultaneously strengthening the sovereignty of the shared data. Sovereignty and trust represent one of the core pillars on which a data space operates.

This section addresses aspects of sovereignty and trust through the following building blocks.

- **Identity and attestation management.** It specifies the mechanisms required for managing identities and other attestations within the data space. It enables participants to present, verify, store, and exchange attestations in an environment that builds trust and significantly facilitates the onboarding process by ensuring the data sovereignty. The purpose of this building block is multifaceted; it proposes standards for managing evidence of participant identity, explains how attestations are validated within the ecosystem, and details the protocols and standards related to sovereignty and trust. This block includes a strong automation component, and its protocols and standards must be captured in a machine-readable rulebook to support and streamline compliance and verification workflows, credential exchange, and other attestations. Some of the standards covered in this section are ISO/IEC 17000:2020: Conformity Assessment and W3C VC Verifiable Credentials Data Model v2.0.
- **Trust framework.** This building block defines the elements directly related to trust within the data space. Trust is essential in a data space because it not only strengthens the technical capabilities of identity and attestation management but also accelerates decision-making, data exchange, and the overall growth and expansion of the data space. The purpose of this block is to establish and define the technical components of the trust principles applicable to the data space, as well as the roles and responsibilities needed to ensure continued operation in a trustworthy environment. In this case, the CRMDS will be strongly supported by the Gaia-X Trust Framework (Gaia-X, 2022).
- **Access usage policies enforcement.** It defines the technical mechanisms and capabilities needed to apply and enforce access, usage, and consent policies within a data space. Its core objective is to enable controlled and sovereign data sharing by ensuring that policies, whether defined at the data space level or per transaction, can be interpreted and enforced technically during data exchanges.

Access control policies determine who can access data and under what conditions, usage control policies specify what actions are permitted once access is granted (e.g., analysis, modification, sharing), and consent management policies handle the issuance, validation, and revocation of explicit consent, particularly for personal data. This building block recommends interoperable policy representations, often using ODRL, so policies can be uniformly understood and executed across participant systems. It also suggests architectural components such as Policy Enforcement Points (PEP), Policy Decision Points (PDP), and Policy Administration Points (PAP) to structure enforcement workflows.

The implementation of the technical standards defined in this building block will enable comprehensive monitoring of activities within the data space, both those related to participant processes (such as onboarding, credential management, and attestations) and those affecting shared datasets. This will facilitate the creation of metrics and audit logs that further strengthen the trust layer of the ecosystem.

The *value creation services building block* provides a detailed list of performance monitoring and metrics capabilities.

4.2.3 Data value creation building block

A core objective of a data space is to enable value creation among its participants through data sharing. To achieve this, participants must be able to identify and understand the available data, services and offerings.

This section provides the technical foundations needed to support that goal, structured across three building blocks.

- **Data, services and offering descriptions.** This building block defines the specifications, standards, tools, and processes needed to create high-quality metadata for data products, technical services, and combined offerings within a data space. Quality metadata is essential for making resources discoverable, interoperable, and usable by both humans and machines, forming a foundational element of efficient data sharing ecosystems. This block emphasises adherence to FAIR principles and encourages standardisation using widely accepted vocabularies such as the DCAT. It also covers how related policies and constraints, such as access and usage terms, can be incorporated using languages like ODRL, and describes how metadata processes (creation, validation, and maintenance) support the needs of providers, recipients, intermediaries, and governance authorities. The scope includes detailed treatment of dataset descriptions, service descriptions, and offerings descriptions, which may include pricing, licensing, lifecycle information, and rights and obligations, all organised

into catalogues for effective discovery and consumption. This building block connects closely with publication and discovery, interoperability, and other data value creation enablers. The following figure summarizes the core elements involved in this building block and how they articulate through a data catalogue.

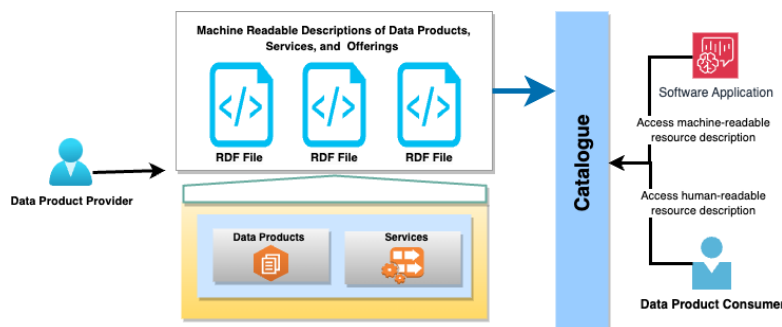


Figure 4-8. Data catalogue: descriptions, data products, and services

- Publication and discovery.** This block enables the provisioning and discovery of data, services, and other offerings within a data space. It supports the publication of metadata descriptions created by providers (representing data and service offerings) into a catalogue where they are managed throughout their lifecycle, that is, published, updated, and removed. Consumers can query this catalogue to find offerings that best match their business and technical needs and request access to them. Through this technical building block, the data space benefits from:
 - ✓ Management of offerings, allowing providers to make their metadata visible and control access.
 - ✓ Discovery of offerings, enabling consumers to search and identify suitable data or services.
 - ✓ Lifecycle support, helping providers maintain offerings from publication through to removal.
 - ✓ Flexible architecture, permitting both centralised and decentralised catalogue models and supporting dynamic access control.

The building block is closely linked to data, services, and offerings descriptions, which define what is published; it also aligns with the regulatory compliance block and with the next block, value creation services as well, as it supports discovery in ways that respect policy, law, and value generation. Overall, this block plays a central role in ensuring that participants can find and connect to relevant data and services across a data space, facilitating efficient and FAIR-aligned data sharing.

- **Value creation services.** Lastly, this block defines the technical aspects of services that enable participants to generate real value from the data shared within a data space. This building block complements core infrastructure services (such as federation and participant agent services) by offering functional capabilities that go beyond basic data exchange and governance.

These services can be divided into different categories according to the purpose within the data space. Each of the following categories lists some of the services that can be implemented:

- ✓ **Core services.** These are the minimum capabilities a data space must have to operate efficiently. The core services are transversal and are not linked to specific datasets. Among others, they should provide:
 - Data visualisation.
 - Data quality management, assessment and validation.
 - Enablers for automatic compliance.
 - Security.
 - Monitoring and reporting.
- ✓ **Data handling services.** Acting directly over specific datasets to facilitate their access and use:
 - Data filtering and packaging allowing the combination of different datasets.
 - Data processing and transformation.
 - Data delivery, that guarantees access in an efficient way.
- ✓ **Value-added services (VAS).** Given that they add an extra value to standard data or services, their capabilities should be disclosed in the provider's offer. These services include, but are not limited to:
 - Data enrichment, by combining multiple sources of information and/or using domain-specific techniques.
 - Collaborative data analytics.
 - Machine Learning (ML).
 - Artificial Intelligence (AI).
 - Training and education.
 - On-demand services and customisable workflows.
- ✓ **Infrastructure integration services.** Enable connection to external infrastructures for processing, storage, and data collection.
 - Infrastructure catalogue.
 - Infrastructure orchestration and load balancing.
 - Cloud/edge/HPC provisioning and resource scaling.

- ✓ Application integration services. Enable the connection to external applications to be deployed in the data space, allowing a seamless integration.
 - AI integration services.
 - Digital twins.
 - Vertical industry solutions applied to specific sectors.
- ✓ Business enablement services. Support business models and transactions within the data space.
 - Billing and automated payments.
 - Smart contracts for secure, automated agreements.
 - Certifications for data/service quality and compliance.

The most direct example of value creation in a data space is the *marketplace*. It enables the offering of products and services that address one or several use cases and can be complemented by the value-added services described above.

In addition, a data space marketplace always works based on the operational trust rules that govern the ecosystem, ensuring data sovereignty and providing the technical and legal tools that allow participants to negotiate contracts securely, clearly, and reliably.

In summary, all technical, governance, and legal activities within the data space must align with this building block so that the data space generates intrinsic value through the process of federated data transactions in compliance with:

- **Common technical and semantic standards.** Shared data models, taxonomies, and controlled vocabularies are determined to ensure consistent interpretation of data. Interchangeable formats, exchange protocols, and compatibility with federation mechanisms are also specified.
- **Data exchange interfaces and APIs.** The functional, security, and performance requirements of APIs are defined, including authentication and data validation. This dimension ensures that providers and consumers interact in a fully interoperable manner.
- **Digital identity, authentication and authorisation.** Identity management services are implemented to enable secure, decentralised authentication of participants. Authorisation must be expressive and granular, incorporating policies on use, consent and control by the data provider.
- **Traceability, monitoring and auditing.** This includes mechanisms that enable all activities within the data space to be recorded, verifiable evidence of the

exchange to be generated, policy compliance to be monitored and internal or external audits to be carried out. This is essential to guarantee data sovereignty.

- **Availability and service quality requirements.** Service levels, response times, incident management procedures, technical redundancies, and failover protocols are detailed. All of this ensures the operational continuity of the environment.
- **Connection to federated ecosystem services.** The functions necessary to interact with federated catalogues, discovery services, or trust verification mechanisms between data spaces are described.
- **Conditions for accessing and using data.** The criteria for becoming a data provider, consumer or intermediary are established, as well as the permissions and obligations for use. This includes the possibility of establishing levels of access, specific conditions for certain sensitive data sets and explicit commitments regarding the prohibition of unauthorised use.
- **Intellectual property and licensing rules.** The licensing terms applicable to data, metadata and associated services are described, ensuring that original ownership is respected and that there is clarity regarding the use of shared data.
- **Economic and compensation models.** As defined in the business model based on the CRMsDS use case scenario.
- **Quality and accountability commitments.** Minimum data quality parameters are defined, as well as responsibilities in the event of errors or incidents in data supply services.
- **Conflict resolution mechanisms.** Formal mediation or arbitration procedures are established to deal with discrepancies in a manner that does not disrupt the data space ecosystem.

5 Data space compliance overview and communication with stakeholders

5.1 Governance control and audit mechanisms

Governance control within the CRMs Data Space is grounded in a federated governance framework that combines organisational, operational and technical oversight while preserving data sovereignty. Control mechanisms will be exercised through clearly defined roles, responsibilities and decision-making structures, with the governance authority holding the mandate to enact and update governance rules based on agreements among participants.

At this stage of the project, an initial governance model is proposed with the aim of guiding progress through the early phases in a structured manner, ensuring that the foundations laid now will support the establishment of a definitive governance model as the various components of the project evolve. The elements to be defined to continue strengthening and expanding the governance model, include:

- **Participant onboarding:** Pre-conditions will be established to determine the eligibility of candidate participants seeking access to the data space, together with post-conditions that ensure their technical and interoperability readiness.
- **Access and usage policies:** The policies governing access to and use of data within the data space will be defined, ensuring alignment with the future development of a business model applicable to the data space.
- **Alignment of the MVDS with project requirements:** In this phase, the governance authority will centralise coordination and oversight functions, with the intention of later progressing toward more advanced governance mechanisms, including the delegation of functions, conflict-resolution procedures, and distributed decision-making. To support this evolution, inputs and results from the various work packages, along with projections regarding the evolution of the ecosystem, will be carefully analysed and incorporated.

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Compliance with these governance rules is supported by policy enforcement mechanisms embedded in data transactions, where access, usage and sharing conditions are formally defined and machine-enforceable. Auditability will be ensured through traceability, logging and signature mechanisms that record policy application, contract negotiation and data exchange events, enabling transparent verification of compliance and facilitating internal and external audits without centralising data ownership.

5.2 Data space compliance indicators

A set of governance, quality and interoperability indicators aligned with European Data Space principles will be defined to monitor compliance within the CRMs Data Space. These combined indicators will reflect adherence to data sovereignty requirements, FAIR data management practices, defined semantic and technical standards, and compliance with applicable regulatory frameworks.

This set will include additional indicators regarding data quality and integrity, including validation results, metadata completeness, provenance and consistency checks performed during onboarding and lifecycle management.

Together, these qualitative and procedural indicators provide measurable evidence that participants, data assets and services operate in accordance with the governance framework, supporting trustworthy data exchange and continuous monitoring of the ecosystem's conformity.

5.3 Incident management and corrective measures

Incident management in the CRMs Data Space will be integrated into the governance and operational framework to ensure resilience, continuity, and regulatory compliance. Incidents may arise from technical failures, data quality issues, security events, or policy non-compliance and are identified through continuous monitoring, logging and participant feedback, among others.

Corrective measures will be coordinated through predefined procedures involving relevant roles, such as the data space operator, intermediary, or governance authority, depending on the nature of the incident. Actions may include technical remediation, policy enforcement adjustments, data correction or revalidation, and updates to governance rules or operational procedures.

This structured approach ensures that incidents are addressed consistently, lessons learned are incorporated into governance reviews, and the integrity and trustworthiness of the data space are maintained over time.

6 Key success factors & Dependencies with other tasks

6.1 Awareness of governance framework as a starting point

As mentioned above, it must be remembered that the governance framework defined at this stage are preliminary and may be subject to changes as the project progresses. As development advances and requirements evolve, adjustments may be necessary to ensure alignment with project needs, data availability and technical constraints. The governance framework described in this deliverable should therefore be considered a first version rather than a final specification.

6.2 Dependencies with other project tasks

A critical factor in the design and evolution of the governance framework is the nature of its dependencies on other project tasks, whether these are unidirectional or bidirectional.

The following figure provides a diagram of these dependencies:

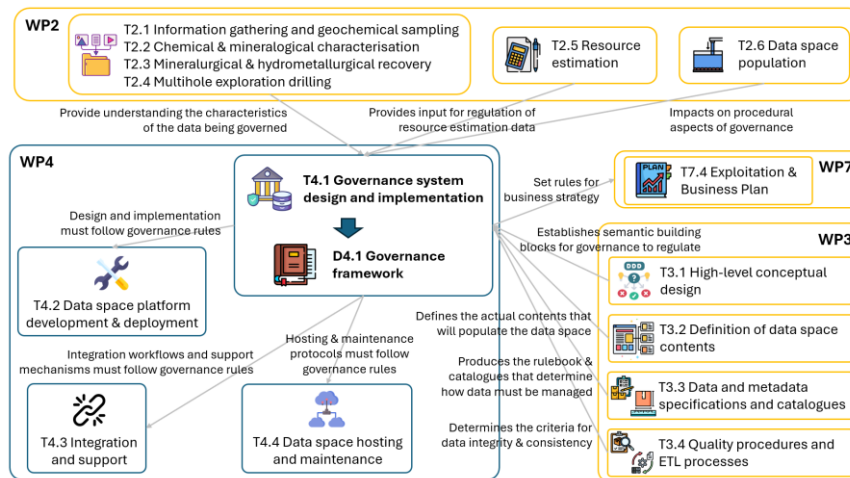


Figure 6-1. Dependencies of task T4.1 with other tasks

These dependencies are as follows:

- Consolidated tasks T2.1 (Information gathering and geochemical sampling), T2.2 (Chemical & mineralogical characterisation), T2.3 (Mineralurgical & hydrometallurgical recovery) and (T2.4 Multihole exploration drilling) → T4.1:

- ✓ T4.1 depends indirectly on the outputs of Tasks 2.1–2.4 because understanding governance requirements requires understanding the characteristics of the data being governed. Field data collection, chemical and mineralogical analyses, recovery results and drilling outputs determine the complexity, sensitivity and heterogeneity of the datasets that the governance model must support. These data characteristics influence crucial governance dimensions such as access control, confidentiality rules, validation requirements, sovereignty mechanisms and data provider obligations. Governance cannot be designed in the abstract, but must reflect real data flows, data types and data quality considerations derived from WP2.
 - ✓ Tasks 2.1-2.4 are included in *WP2 - Data acquisition from closed facilities* and their objective is to acquire and analyse all empirical data describing the closed extractive facilities, including geochemical sampling, chemical analyses, recovery tests and subsurface drilling. They seek to generate a robust scientific dataset that captures the physical, chemical and geological attributes of the studied sites. This comprehensive dataset provides the technical grounding for the later layers of modelling, catalogue definition, governance and platform deployment, ensuring that the data space is built around authentic, well-characterised data assets.
- **T2.5 (Resource estimation) → T4.1:**
 - ✓ T4.1 also depends on T2.5, which generates resource estimation outputs such as block models, grades and 3D representations. These data products carry specific governance implications related to intellectual property, confidentiality, accuracy and quality assurance. Because resource estimates may be sensitive, economically relevant, or derived from proprietary modelling software, governance rules must incorporate safeguards for data sovereignty, access rights, permissible reuse and roles of the different organisations contributing the data. Thus, T2.5 provides essential input for determining the governance policies that will regulate how resource estimations are shared and controlled inside the data space.
 - ✓ T2.5 is included in *WP2 - Data acquisition from closed facilities* and its objective is to perform comprehensive resource estimation using the RecMin program, producing geological models, geostatistical analyses and block estimations for the targeted critical raw materials. This task translates raw sampling and drilling data into a structured understanding of resource distribution, forming a critical part of the data space's strategic value. Because of its importance and potential sensitivity, this

information requires precise governance rules, making T2.5 a relevant dependency for T4.1.

- **T2.6 (Data space population) → T4.1:**

- ✓ T4.1 depends on T2.6 because the task operationalises how data flows into the data space, making it a key input for designing governance processes such as validation workflows, roles and responsibilities, traceability requirements and compliance monitoring. Understanding how partners will upload and manage data is essential for defining who is allowed to contribute data, under what conditions and with what oversight. T2.6 therefore influences procedural aspects of governance, especially around lifecycle management, quality enforcement and system responsibilities.
- ✓ T2.5 is included in *WP2 - Data acquisition from closed facilities* and its objective is to perform comprehensive resource estimation using the RecMin program, producing geological models, geostatistical analyses and block estimations for the targeted critical raw materials. This task translates raw sampling and drilling data into a structured understanding of resource distribution, forming a critical part of the data space's strategic value. Because of its importance and potential sensitivity, this information requires precise governance rules, making T2.5 a relevant dependency for T4.1.

- **T3.1 (High-level conceptual design) → T4.1:**

- ✓ T4.1 depends on T3.1 because governance rules require a clear understanding of the fundamental components of the data model before defining policies, obligations and controls. T3.1 establishes the semantic building blocks that governance must regulate (phenomena, properties, attributes, inheritance structures, relations, restrictions, observation and measurement concepts and metadata profiles). Without this conceptual foundation, governance cannot specify rules for how data entities must be managed, validated, accessed, or interpreted. In other words, the governance framework can only function if the conceptual model underneath is fully defined, stable and standardised.
- ✓ T3.1 is included in *WP3 – Development of the data catalogue* and its objective is to analyse applicable European norms and standards and produce a high-level conceptual design of the data space, defining the essential elements that structure all data representations. This includes identifying the semantic architecture, conceptual entities, metadata profiles and organising principles that will later be operationalised across the platform. The task ensures that the consortium has a shared conceptual understanding of how data should be represented,

interpreted and linked, forming the foundation needed for governance enforcement.

- **T3.2 (Definition of data space contents) → T4.1:**
 - ✓ T4.1 depends on T3.2 because this task defines the actual contents that will populate the data space, including the characterisation of phenomena, attributes, relations, controlled vocabulary and multimedia elements. Governance policies must be tailored to regulate these specific data contents, determining who can contribute, modify, access, or disseminate each type of information. Without T3.2, governance would lack the necessary understanding of what data assets exist in the ecosystem, what sensitivities they may carry, what interoperability constraints apply and what obligations must guide their use. As a result, governance rules must be derived from the precise content structures defined in T3.2.
 - ✓ T3.2 is included in *WP3 – Development of the data catalogue* and its objective is to develop the complete semantic definition of the data space contents, characterising all phenomena and their associated properties, relations, restrictions and terminological controls. This task integrates the empirical data collected in WP2 with the semantic foundations defined in T3.1, transforming raw information into structured, interoperable data space elements. Its purpose is to create a comprehensive and standardised representation of all relevant geological, mineral, environmental and contextual datasets so that they can be managed consistently across the platform.
- **T3.3 (Data and metadata specifications and catalogues) → T4.1:**
 - ✓ T4.1 heavily depends on T3.3 because governance must enforce the rules, constraints and interoperability standards defined in the data and metadata specifications. T3.3 produces the Rulebook and the detailed catalogues that determine how heterogeneous data must be structured, harmonised and exchanged. Governance cannot be defined without these specifications, because policies related to data quality, access rights, lifecycle management, compliance and interoperability hinge on the metadata obligations and standardised APIs provided in this task. In effect, T3.3 provides the operational semantics that governance structures must police.
 - ✓ T3.3 is included in *WP3 – Development of the data catalogue* and its objective is to develop the rulebook and detailed semantic and technical catalogues that enable interoperability across systems and stakeholders. This includes defining the semantic data model, the tools and methods for extracting and harmonising data, compliance with privacy and

security requirements and the open APIs required to implement the core services of the ecosystem. The task creates the formalised, standardised specifications that enable the data space to function as a coherent system, supporting interoperability and trustworthy data exchange.

- **T3.4 (Quality procedures & ETL processes) → T4.1:**
 - ✓ T4.1 depends on T3.4 because the governance framework must incorporate and enforce the quality assurance (QA) and quality control (QC) policies defined in this task. Governance is responsible for regulating how data is validated, transformed, ingested, monitored and corrected during its lifecycle in the data space. The QA/QC rules and ETL processes defined in T3.4 determine the criteria for data integrity, consistency, provenance and quality thresholds, elements that governance must codify and formalise into binding procedures.
Without T3.4, governance would lack a concrete basis for establishing compliance and quality-related obligations.
 - ✓ T3.4 is included in *WP3 – Development of the data catalogue* and its objective is to define the project's QA/QC policy and automate its application during the ETL process. This includes establishing the catalogue of requirements and mechanisms to guarantee data quality, developing tools to automate field data capture and ensuring data integrity and consistency throughout the data lifecycle. By setting measurable and enforceable quality standards, T3.4 equips the platform with robust procedures that governance will ultimately formalise and oversee in T4.1.
- **T4.1 → T4.2 (Data space platform development and deployment):**
 - ✓ T4.2 depends on T4.1 because the data space platform must be designed and implemented in alignment with the governance rules, data sovereignty principles, organisational roles and compliance mechanisms defined in the governance framework. T4.1 sets out how data providers, consumers and managers must interact with the platform and these rules drive key architectural decisions regarding access control, identity management, trust services, security enforcement and interoperability protocols.
Without T4.1, T4.2 would lack the policy requirements that must be built into the technical system, making it impossible to implement a platform that properly enforces responsibilities, restrictions and regulatory obligations.
 - ✓ The objective of T4.2 is to develop and deploy the data space platform using state-of-the-art technologies aligned with European frameworks such as EDC, IDSA DSP and Gaia-X. This includes ensuring scalability,

security, interoperability and seamless integration with existing systems. The task focuses on building the technical backbone of the data space and therefore relies on D4.1 to ensure that its implementation is aligned with the rules, roles and governance requirements that the platform must enforce.

- **T4.1 → T4.3 (Integration and support):**

- ✓ T4.3 depends on T4.1 because integration workflows and support mechanisms must operate in accordance with the governance rules defined in T4.1. These governance rules determine the permissions for connecting external systems, the procedures for validating imported data, the responsibilities of data providers and the workflows for resolving issues or enforcing compliance. Support processes such as troubleshooting, user onboarding and governance monitoring also require the organisational structures and policy definitions established in T4.1.

Without these governance conditions, T4.3 would lack the necessary framework to perform integration safely, consistently and in line with regulatory principles.

- ✓ The objective of T4.3 is to integrate diverse data sources, systems and tools into the data space platform and provide ongoing user support, troubleshooting, governance compliance assistance and performance assurance. This task ensures that the data space becomes a functional, collaborative environment where different stakeholders can interact and exchange data reliably. To do so, it must follow the governance rules set out in T4.1, which define how integration must occur and what oversight mechanisms must be applied.

- **T4.1 → T4.4 (Data space platform maintenance):**

- ✓ T4.4 depends on T4.1 because hosting and maintenance protocols (including update cycles, logging requirements, backup procedures, operational monitoring and threat mitigation) must comply with the governance rules established in the governance framework. Governance elements such as data sovereignty, access restrictions, auditability and lifecycle rules directly influence how the platform must be hosted and maintained.

For example, sovereignty principles may require certain datasets to remain in specific infrastructures; access policies determine how systems must authenticate users; and governance-level compliance rules define how incidents and security breaches must be managed.

Thus, D4.1 sets the constraints and obligations that T4.4 must incorporate into its operational procedures.

- ✓ The objective of T4.4 is to host and maintain the data space platform using a hybrid on-premise/cloud model that ensures availability, security and optimal performance over time. This includes system updates, performance monitoring, backups, data security enforcement and threat mitigation. T4.4 therefore must implement procedures that reflect the governance rules defined in D4.1, ensuring that operational maintenance aligns with the project's regulatory, organisational and interoperability principles.
- **T4.1 → T7.4 (Exploitation and Business Plan):**
 - ✓ T7.4 depends on T4.1 because the governance framework developed in T4.1 defines the organisational rules, roles, responsibilities, compliance mechanisms, data sovereignty principles, collaboration structures and regulatory guidelines under which the data space will operate. These governance mechanisms determine how the ecosystem can be exploited after the project ends: for example, who can own, manage, or commercialise services; what access models or licensing frameworks are possible; how data sovereignty and FAIR principles must be enforced; and what obligations a future spin-off or consortium-led entity must respect. Without the governance framework defined in T4.1, it would be impossible for T7.4 to construct a credible, compliant business plan, since exploitation must reflect the governance model under which the data space will continue to function.
 - ✓ The objective of T7.4 is to develop a Business Plan that implements the project's exploitation strategy, potentially through a new spin-off involving all partners proportionally to their contributions. This plan must define the operational, financial and organisational pathways for sustaining the data space, offering consultancy services, providing user-friendly access to resource estimation and business solutions and continuing development of the data space ecosystem. Because all these exploitation pathways must respect the governance rules established in T4.1 (including roles, responsibilities, sovereignty and compliance) the outputs of T4.1 serve as an essential foundation for T7.4.
 - ✓ The dependency relationship between T4.1 and T7.4 is explained in further detail below.

6.2.1 Alignment of governance framework and business plan

The relationship between the governance framework and business plan is one of the most structurally important inter-work-package relationships in the entire project, because the governance framework effectively defines the legal, organisational, operational and regulatory boundaries within which all exploitation activities envisioned

in the business plan must occur. Governance sets the rules of the game, so to speak, and the business plan must strategize within those rules.

The governance framework establishes a governance system for the data space that includes the organisational structure, the authority and responsibilities of different stakeholders, the policy and regulatory framework and the mechanisms for data sovereignty, compliance, interoperability and collaboration across partners. These components are not merely administrative, since they directly determine what types of exploitation pathways are possible, permissible, or sustainable. For example, if governance rules specify that certain data assets retain strict sovereignty under the control of specific partners, then the business plan must define exploitation mechanisms that honour that sovereignty, potentially limiting the forms of commercialisation or structuring them under controlled access models. Similarly, if the governance system defines mandatory FAIR principles, interoperability obligations or security requirements, the exploitation strategy must ensure that all post-project services and operations remain compliant.

In practical terms, governance determines how future services may be delivered, which organisation(s) may provide them, what licensing or access models are legally and operationally feasible and what obligations accompany the reuse or redistribution of data. For instance, the governance rules clarify whether a spin-off, a possibility that the business plan must consider, can act as the main operational entity, how ownership and IPR are handled among partners, how revenues or operational costs could be shared and what compliance responsibilities (security, privacy, interoperability) the spin-off must assume. Without the governance framework, the exploitation strategy would lack legal clarity, organisational grounding and operational feasibility.

Moreover, governance provides stability, which is essential for exploitation. Long-term business planning cannot rely on ad-hoc agreements; it requires a well-defined system of authority, decision-making processes, enforcement mechanisms and channels for stakeholder interaction, all of which are delivered by the governance framework. Exploitation activities such as consultancy services, training programs, resource-estimation services, new data space developments, or the maintenance of the platform's operational environment must align with the governance constraints and take advantage of the governance structures that T4.1 builds.

The governance framework also defines the collaboration mechanisms between partners, including procedures for communication and coordination. The business plan must embed these mechanisms into its business models, especially if the proposed spin-off will rely on contributions, expertise, or data streams from multiple partners. Governance defines roles (data providers, data consumers, data space managers, service providers, etc.) that naturally translate into operational roles within the business plan, which interprets these roles to determine who will be responsible for which

components of the service offering, what their liabilities are and how value will be created and distributed.

Finally, governance embeds regulatory and policy compliance, particularly related to the European Data Space principles and the FAIR framework. This has a profound impact on the business plan: any exploitation model must demonstrate compliance with the policy environment the project is meant to support and contribute to. The business plan must show continuity with the governance system, because exploitation of a data space ecosystem is inseparable from respecting the regulatory context under which it was built.

In summary, the governance framework is a foundational prerequisite for the business plan. Governance functions as the normative, organisational and regulatory bedrock; exploitation is the strategy built on top of that bedrock. The business plan cannot propose credible, sustainable exploitation pathways (nor design viable service models, spin-off structures, or commercialisation mechanisms) without grounding every aspect of the business plan in the governance rules, role structures, sovereignty principles, compliance obligations and collaboration mechanisms defined in the governance framework.

The table below summarizes the main points to be considered when aligning the governance framework and the business plan, as well as some examples for illustrative purposes only.

Table 6-1. Integration of data space governance framework and business plan

Governance component	Impact on business plan	Example
Organisational structure & Governance authority	Determines the management model for exploitation, including roles in a potential spin-off and responsibility allocation among partners.	If governance assigns SATEC as platform manager, the business plan must reflect that the spin-off or consortium body relies on SATEC for operational oversight.
Roles & Responsibilities	Defines the operational roles in the exploitation model and who performs which services.	If GIG-PIB is the official resource estimation expert, the business plan must position GIG-PIB as the lead in future consultancy services.
Policy & Regulatory framework	Ensures exploitation strategies (especially monetisation) respect mandatory regulatory conditions such as interoperability, accessibility, and sovereignty.	The business plan cannot propose exclusive licensing models that violate FAIR or Gaia-X principles; it may offer tiered access levels instead.

Governance component	Impact on business plan	Example
Data sovereignty	Defines what can be commercialised, what must remain under owner control, and what access or use rights a spin-off can offer.	A partner owning sensitive drill-hole data may allow only aggregated resource estimation results to be commercialised.
Collaboration & Decision-making	Shapes how the future spin-off or governance body makes strategic decisions on upgrades, pricing, partner participation, etc.	If governance requires consensus for major decisions, the business plan must embed this into its operational rules or bylaws.
Security & Privacy	Determines which services can be offered externally and which require restricted access or increased security protocols.	Resource estimation consultancy may be public; access to raw geochemical data may require strict authentication and encrypted channels.
Interoperability	Influence which external systems can be integrated into the commercial service model and which require certification or compliance steps.	The business plan may include a paid “integration support service” helping external companies map their datasets to the project’s metadata standards.
Access control & Rights Management	Defines the pricing, licensing, and service-access models possible in the exploitation plan.	A “freemium” access model: basic facility information is public, while advanced data analytics (AI classification, resource estimation) are premium services.
Communication & Collaboration	Determines how the business entity interacts with the consortium and stakeholders after project completion.	Governance may require periodic reports from the spin-off to the original consortium, shaping the business plan’s reporting structure.
Auditability & Traceability	Inform the business model around compliance services or data integrity guarantees.	The business plan may offer “audit-ready data exports” or “compliance verification services” to industrial users.
Quality Assurance	Impact service guarantees, SLAs, and the reliability of outputs offered by the business model.	The business plan can advertise QA-certified data products (e.g., validated resource estimation models) as premium deliverables.
Sustainability & Long-term governance	Determine how the business plan ensures long-term operation of the data space beyond the project.	Governance may require continuous reinvestment in platform updates; the business plan must allocate part of revenues to maintenance.

6.2.2 Sustainability and evolution of CRMDS within the business plan and the governance framework

While the governance framework establishes the structural, legal and organisational conditions under which the data space operates, the responsibility for ensuring long-term sustainability and evolution rests primarily with the business plan. Sustainability is fundamentally an economic, operational and market-driven objective, and therefore depends on the business plan's ability to articulate viable revenue models, service portfolios, cost structures, reinvestment strategies and growth pathways. The business plan must demonstrate how the operational entity will generate sufficient value to maintain and expand the data space beyond the project lifetime. This includes defining strategies for recurring income streams, customer onboarding, partnerships, maintenance funding and service innovation. Governance provides the rules, but it is the business plan that transforms them into a self-sustaining operational reality.

Nonetheless, the governance framework plays an essential complementary role by providing the institutional stability and procedures that make long-term business planning credible. Governance does not deliver sustainability itself but creates the environment in which sustainable business models can be conceived and executed. It does so by defining enduring roles, decision-making processes, update procedures and compliance obligations that the business plan must rely on to design realistic long-term strategies. Governance ensures continuity and legal certainty, allowing the business plan to focus on market orientation, service evolution and financial resilience.

Because the data space must adapt to evolving technologies, partner needs and regulatory developments, both business plan and governance must address evolution, yet from different angles. The business plan explains why change is necessary and which service, organisational or market adaptations will be pursued, while governance defines how changes can occur (e.g., update cycles, partner onboarding or policy revision protocols). This distinction ensures that evolution is both strategically justified and procedurally legitimate. In practice, the business plan should outline growth scenarios, service diversification pathways and mechanisms for scaling the ecosystem, while operating within the framework defined by governance.

An important issue that lies with the business plan's strategic foresight is anticipating and responding to future EU regulatory developments, such as those stemming from the Data Governance Act or the Critical Raw Materials Act. This is especially important considering that the CRMDS project intends to provide an overarching Common European Data Space framework for CRMs.

The business plan must explain how compliance requirements will shape commercial offerings, partnership models and data-access services, ensuring that exploitation activities remain competitive and lawful within the evolving European data economy.

Governance contributes by defining the compliance principles and auditability rules that form the foundation for regulatory alignment, but it is the business plan that operationalises these into viable services, transparent value propositions and long-term business opportunities fully aligned with EU data policies.

7 Conclusions & Next steps

This deliverable has established the first version of a governance framework for the CRMs Data Space, providing a structured and coherent foundation for the development of a Minimum Viable Data Space (MVDS) aligned with European Data Space principles. The framework consolidates organisational, legal, semantic and technical dimensions into a unified governance model that enables trusted, sovereign and interoperable data sharing among heterogeneous stakeholders in the CRMs domain.

From a conceptual perspective, the governance framework clarifies how a federated data space can operate without centralising data or decision-making authority, while still ensuring coordination, compliance and accountability. By explicitly adopting a federated and unincorporated data space model, the framework respects the autonomy, regulatory diversity and commercial sensitivities of participating organisations, which is a prerequisite for sustainable collaboration in the CRMs ecosystem. Governance is therefore positioned not as much as a mechanism of control, but as an enabler of cooperation, trust and long-term viability.

A key contribution of this deliverable is the definition of governance roles, responsibilities and interaction mechanisms tailored to the MVDS. The clear distinction between roles such as data owner, data provider, data operator, data space administrator and governance authority establishes predictable responsibilities across the data lifecycle, from data creation and onboarding to access, use and monitoring. This role-based approach supports accountability and transparency while remaining flexible enough to accommodate different organisational arrangements and future participants.

The framework also defines how policies, contracts and technical enforcement mechanisms jointly regulate data exchange. By linking governance rules to machine-enforceable policies and contract-based data transactions, the framework ensures that data sovereignty principles are operationalised rather than remaining purely declarative. This alignment between organisational governance and technical enforcement is essential for ensuring compliance with FAIR principles, European interoperability standards and emerging EU regulatory instruments such as the Data Governance Act and the Critical Raw Materials Act.

Another important outcome is the explicit positioning of governance as a bridge between the semantic and technical foundations defined in WP3 (data catalogue) and the platform implementation activities in WP4 (MVDS). Governance rules are shown to depend directly on the nature, structure and sensitivity of the data defined in upstream tasks, while at the same time constraining and guiding downstream platform design, integration, support and maintenance activities. In this sense, the governance

framework functions as a connective layer that translates conceptual models and data specifications into operational rules and enforceable practices.

It is also important to emphasise that the governance framework presented in this deliverable is intentionally preliminary. At this stage of the project, governance must remain adaptable, as data availability, technical implementation choices, regulatory interpretations and stakeholder requirements will continue to evolve. Rather than providing a rigid or final specification, this deliverable establishes a governance baseline that can be iteratively refined as the MVDS is implemented, tested and expanded. This evolutionary approach is consistent with both federated data space principles and the incremental nature of MVDS development.

The immediate next steps following this deliverable focus on operationalisation and alignment rather than redesign. Within WP4, the governance rules defined here will be progressively embedded into the technical architecture of the data space platform developed under Task 4.2. This includes implementing identity and access management, policy enforcement, onboarding workflows, logging and audit mechanisms in line with the governance framework.

In parallel, integration and support activities in Task 4.3 will apply the governance rules to real data flows, participant onboarding and issue resolution processes. This practical application will provide valuable feedback on the clarity, feasibility and completeness of the governance model, allowing targeted adjustments where necessary. Platform maintenance activities in Task 4.4 will further operationalise governance requirements related to availability, security, monitoring and incident management.

Beyond WP4, close coordination with WP7 (communication and dissemination) is essential to ensure alignment between governance and exploitation planning. As the Business Plan is developed, the governance framework will serve as the normative and organisational reference that constrains and enables future sustainability and exploitation strategies. Any proposed post-project operational model, including the potential creation of a spin-off or long-term service entity, must be consistent with the roles, decision-making mechanisms, sovereignty principles and compliance obligations defined in this framework.

Finally, periodic governance reviews should be scheduled as the project progresses. These reviews will assess whether the framework remains aligned with evolving technical implementations, stakeholder needs and regulatory developments at EU level. Such reviews will ensure that the CRMs Data Space governance remains robust, credible and fit for purpose, supporting both the successful deployment of the MVDS and its long-term evolution as a Common European Data Space for CRMs.

8 Glossary

AI – Artificial Intelligence

API – Application Programming Interface

CRMA – Critical Raw Materials Act

CRMs – Critical Raw Materials

CRMsDS - Critical Raw Materials Data Space

CTIC – Centro Tecnológico de la Información y la Comunicación

DAMA-DMBOK – Data Management Book of Knowledge

DCAT – Data Catalog Vocabulary

DCAT-AP – Data Catalog Vocabulary-Application Profile

DSP – Data Space Protocol

DSSC – Data Spaces Support Centre

EDC – Eclipse Dataspace Components

eIDAS – Electronic Identification, Authentication, and Trust Services

EIF – European Interoperability Framework

EIRA – European Interoperability Reference Architecture

ETL – Extract, Transform, Load

EU – European Union

FAIR – Findable, Accessible, Interoperable and Reusable

GDPR – General Data Protection Regulation

GIG-PIB – Główny Instytut Górnictwa - Państwowy Instytut Badawczy

GTCs – General Terms and Conditions

HPC – High Performance Computing

ICT – Information and Communication Technology

IDS DSP – International Dataspaces Association Data Space Protocol

INSEMEX – Institutul Național de Cercetare - Dezvoltare pentru Securitate Minieră și Protecție Antiexplozivă

INSPIRE – Infrastructure for Spatial Information in Europe

ISO/IEC – International Organization for Standardization / International Electrotechnical Commission

IT – Information Technology

JSON-LD – JavaScript Object Notation for Linked Data

MoUs – Memoranda of Understanding

MVDS – Minimum Viable Data Space

NIFO – National Interoperability Framework Observatory

ODRL – Open Digital Rights Language

OGC – Open Geospatial Consortium

PAP – Policy Administration Points

PDP – Policy Decision Points

PIMS – Personal Information Management Systems

PEP – Policy Enforcement Points

Q/A-Q/C – Quality Assurance - Quality Control

SATEC – Sistemas Avanzados de Tecnología, S.A.

SLAs – Service Level Agreements

THGA – Technische Hochschule Georg Agricola

UBU – Universidad de Burgos

UNFC – United Nations Framework Classification for Resources

UNIOVI – Universidad de Oviedo

VAS – Value-Added Services

VC – Verifiable Credentials

W3C – World Wide Web Consortium

XACML – eXtensible Access Control Markup Language

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